

Does economic strength or political and civil freedom improve international tourism? An econometric analysis

Amrutha Seshagiri, Soudeep Deb, Tirthatanmoy Das

Decision Sciences
Indian Institute of Management Bangalore

How do **economic capacity (ECO)** and **civil liberties and political freedom (CIVIL)** jointly influence inbound tourism, and to what extent can strength in one dimension **compensate for weakness in the other**?

Literature review

Economic conditions

Income, prices, infrastructure and trade significantly influence tourism demand; their effects can be nonlinear and heterogeneous across countries and source markets.

Crouch (1994); Turner & Witt (2001); Eugenio-Martin et al. (2008); Eita et al. (2011); Martins et al. (2017); Yerdelen Tatoğlu et al. (2020)

Freedom & Institutions

Political freedom, civil liberties and institutional quality also influence inbound tourism through perceived risk, mobility and destination attractiveness.

Caruana & Crane (2011); Su & Lin (2014); Saha et al. (2017); Bulut et al. (2020); Ogbuabor et al. (2024); Cao et al. (2025)

Joint Perspective

A smaller literature considers economic and institutional conditions together and suggests that destination strengths may compensate for weaknesses elsewhere.

Kubickova (2016); Demir & Gozgor (2019); Agbola et al. (2020); Corne (2024); Zhao et al. (2025)

Research gap

Existing studies largely estimate **separate or additive effects**. They do not formally quantify how economic capacity and political/civil freedom **interact or substitute for one another** in determining tourism demand.

Research question

How do **economic capacity (ECO)** and **civil liberties and political freedom (CIVIL)** jointly influence inbound tourism, and to what extent can strength in one dimension **compensate for weakness in the other**?

Findings

- 1 Both **economic capacity (ECO)** and **civil/political freedom (CIVIL)** significantly influence international tourism.
- 2 The estimated tourism impact of CIVIL is substantially larger than that of ECO.
- 3 ECO and CIVIL are **partial substitutes**, with an average estimated elasticity of substitution (σ) of approximately 0.51.
- 4 The degree of substitutability varies substantially across countries and economic conditions.

Answer to our research question

Economic and institutional conditions do not influence tourism independently. Their effects depend on one another, and one dimension can partially compensate for weakness in the other—but the extent of compensation is country-specific.

Implications for tourism policy

High substitutability

- ECO and CIVIL can partially compensate for each other.
- Policymakers have greater flexibility in deciding which constraint to address first.
- Targeted improvements in one dimension can support tourism even when the other is relatively weaker.

Low substitutability

- Improvements in one dimension cannot easily compensate for weakness in the other.
- Coordinated economic and institutional improvements become more important.
- Addressing only one constraint may yield limited additional tourism growth.

Policy insight

Tourism policy should depend on a country's existing economic capacity, institutional environment and estimated elasticity of substitution rather than following a universal prescription.

Economic intuition behind substitution

Tourists respond to the full price of travel rather than monetary price alone.

$$P_j^{\text{full}} = \begin{cases} P_j^{\text{money}} & \text{Monetary cost} \\ \lambda T_j & \text{Time cost} \\ \mu R_j & \text{Risk cost} \\ \nu H_j & \text{Hassle cost} \end{cases}$$

Economic Capacity (ECO)

Can reduce:

- infrastructure constraints,
- travel time,
- service limitations,
- inconvenience and hassle.

Civil Liberties and Political Freedom (CIVIL)

Can reduce:

- perceived travel risk,
- institutional uncertainty,
- restrictions on mobility,
- concerns about personal freedoms.

Economic intuition

Because both dimensions can reduce components of the full price of tourism, improvements in one dimension may partially compensate for limitations in the other.

Elasticity of substitution

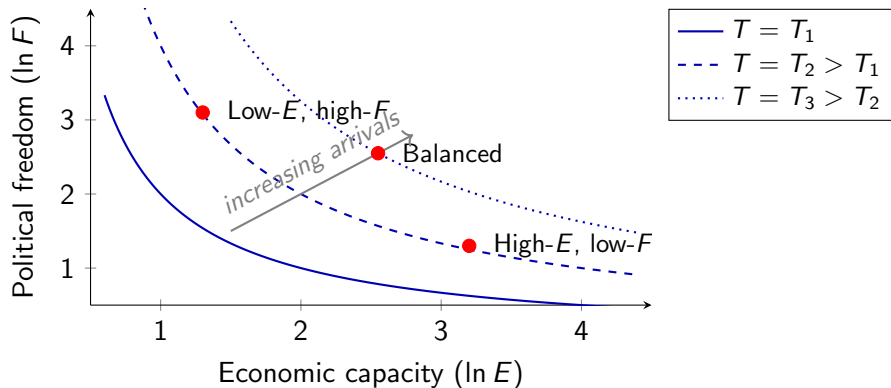


Figure: Iso-arrival contours showing combinations of economic capacity (E) and political freedom (F) associated with equal tourist arrivals. Curvature reflects their degree of substitutability.

Empirical framework

Dataset

- Panel of **171 countries**.
- Period: **2006–2024**.
- Sources: UNWTO, World Development Indicators and Freedom House.

Dimension	Measure
Tourism demand	Inbound tourist arrivals per capita
Economic capacity	Lagged GDP per capita and relative price conditions
Civil Liberties and Political freedom	Freedom Index constructed using PCA of seven Freedom House indicators
Controls	Tourism baseline, continent effects, time trends and COVID indicator

Methodological approach

① Examine information flow

Transfer Entropy is used to examine directional predictive information between tourism demand and macroeconomic/institutional conditions.

② Estimate a flexible tourism demand function

A translog specification captures:

- nonlinear effects,
- interactions between ECO and CIVIL,
- changes in marginal effects across economic and institutional conditions.

③ Estimate elasticity of substitution

Country-specific elasticity measures the degree to which ECO and CIVIL can compensate for one another while maintaining tourism demand.

Transfer Entropy: Examining directional information flow

Need for Transfer Entropy

The regression assumes that **ECO** and **CIVIL** influence **tourism**, but reverse directionality is plausible: tourism may itself affect economic conditions and, over time, institutional openness.

Transfer Entropy (TE)

Transfer entropy is the expected KL divergence between two predictive distributions:

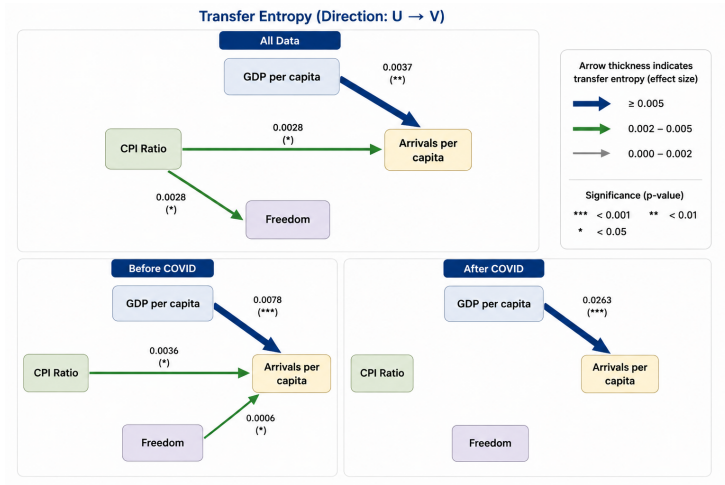
$$TE_{U \rightarrow V} = D_{\text{KL}}(p(v_{t+1} \mid v_t, u_t) \parallel p(v_{t+1} \mid v_t))$$

$$TE_{U \rightarrow V} = \mathbb{E} \left[\log \frac{p(v_{t+1} \mid v_t, u_t)}{p(v_{t+1} \mid v_t)} \right]$$

With U vs. **Without U**

- $TE_{U \rightarrow V} > 0$: U contains additional predictive information about V .
- TE is **directional**: $TE_{U \rightarrow V} \neq TE_{V \rightarrow U}$.

Transfer entropy study - Results



Translog Regression framework

Tourism demand is modeled as a flexible function of the destination's economic and institutional conditions.

$$\begin{aligned}
 \log Y_{j,t} = & \underbrace{\alpha_0}_{\text{Intercept: Baseline level of log tourist arrivals per capita}} + \underbrace{\sum_{i=1}^3 \alpha_i \log X_{i,j,t}}_{\text{Linear effects of the three explanatory variables } (i = 1, 2, 3)} + \underbrace{\delta_1 t}_{\text{Linear time trend}} + \underbrace{\frac{1}{2} \delta_2 t^2}_{\text{Quadratic time trend}} \\
 & + \frac{1}{2} \sum_{i_1=1}^3 \sum_{i_2=1}^3 \beta_{i_1 i_2} \log X_{i_1,j,t} \log X_{i_2,j,t} \quad \left\{ \begin{array}{l} \text{Pairwise interactions among} \\ \text{the three explanatory variables} \\ (i_1, i_2 = 1, 2, 3). \end{array} \right. \\
 & + \underbrace{\sum_{i=1}^3 \gamma_i t \log X_{i,j,t}}_{\text{Interactions of time with each explanatory variable } (i = 1, 2, 3)} + \underbrace{\theta \mathbb{1}(t > 2019)}_{\text{Indicator for the COVID-19 period; equals 1 if } t > 2019, 0 \text{ otherwise (COVID fixed effect)}} \\
 & + \underbrace{\sum_{r=1}^5 \eta_r \mathbb{1}(j \in \mathcal{C}_r)}_{\text{Continent fixed effects. } \mathbb{1}(j \in \mathcal{C}_r) = 1 \text{ if country } j \text{ belongs to continent } r \text{ } (r = 1, \dots, 5), 0 \text{ otherwise}} + \underbrace{\zeta \log Y_j}_{\text{Historical tourist arrivals per capita for country } j} + \underbrace{\epsilon_{j,t}}_{\text{Idiosyncratic error term}}
 \end{aligned}$$

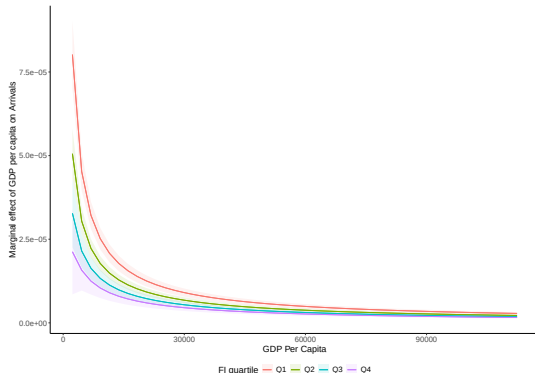
Notation:

- $Y_{j,t}$: Tourist arrivals per capita in country j in period t
- $X_{i,j,t}$: Explanatory variable i for country j in period t ($i = 1, 2, 3$)
- t : Time index (e.g., year or period)
- j : Country index
- $\mathcal{C}_r$: Continent r that country j belongs to ($r = 1, \dots, 5$)
- $\log Y_j$: Historical tourist arrivals per capita for country j
- $\mathbb{1}(\cdot)$: Indicator function (1 if the condition is true, 0 otherwise)
- $\epsilon_{j,t}$: Error term for country j in period t

The translog form allows us to estimate:

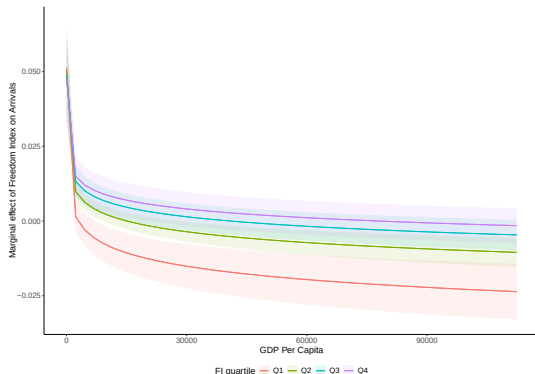
- Marginal effects of ECO and CIVIL
- $\text{ECO} \times \text{CIVIL}$ interactions
- Nonlinear effects
- Country-specific elasticity of substitution.

Marginal returns of economic capacity



- Economic capacity has its greatest tourism impact among lower-income countries.
- The marginal benefit declines as GDP per capita increases.
- Beyond moderate income levels, additional economic capacity contributes less to tourism growth.
- The effect also depends on the country's political and institutional environment.

Marginal returns of civil and political freedom



- Freedom has a positive tourism effect across a wide range of countries.
- Its marginal impact differs with the level of economic development.
- Gains can be particularly important for economically weaker countries that have achieved a basic level of institutional openness.
- The result reinforces the importance of examining ECO and CIVIL jointly.

Elasticity of Substitution and Marginal Effect

Year	Finland		Sweden		Iceland		Croatia		Bangladesh		India	
	EoS	ME(F)	EoS	ME(F)	EoS	ME(F)	EoS	ME(F)	EoS	ME(F)	EoS	ME(F)
2007	3.738	0.003	2.671	0.003	3.284	0.003	-0.449	0.007	-0.087	0.016	-0.474	0.017
2010	2.600	0.003	2.162	0.003	2.821	0.003	-0.811	0.006	-0.019	0.014	-0.484	0.017
2012	2.399	0.003	1.748	0.002	2.265	0.003	-0.453	0.007	-0.065	0.015	-0.334	0.016
2016	1.747	0.003	1.466	0.002	1.472	0.002	-1.919	0.006	0.002	0.012	-0.159	0.015
2018	1.600	0.002	1.461	0.002	1.343	0.001	-1.746	0.006	-0.082	0.010	-0.075	0.014
2021	1.544	0.002	1.390	0.002	1.311	0.001	-3.846	0.005	-1.246	0.007	-0.010	0.013
2024	1.396	0.002	–	–	–	–	–	–	–	–	-0.004	0.011

- Elasticity of substitution varies substantially across countries and over time.
- **Finland, Sweden and Iceland** exhibit relatively high EoS, indicating greater ability of ECO and CIVIL to substitute for one another.
- **Croatia, Bangladesh and India** exhibit low or negative EoS, suggesting stronger need for improvements across both dimensions.

Implication

The same tourism-development strategy will not be equally effective across countries.

Methodological Contribution

- Uses a flexible translog framework to estimate interactions, marginal effects and country-specific elasticity of substitution in tourism demand.

Policy Contribution

- Shows that economic capacity and political freedom should not be viewed as independent determinants of tourism.
- Provides a framework for identifying whether countries can focus on a binding constraint or require coordinated improvements across economic and institutional dimensions.

Thank You

Appendix

Appendix: Transfer Entropy Results

Table 2: Transfer entropy between arrivals per capita and macroeconomic indicators.

V	U	Transfer Entropy	p-value
All data:			
Arrivals per capita	← Lagged GDP per capita	0.0037	0.0067 (**)
Lagged GDP per capita	← Arrivals per capita	0.0009	0.2000
Arrivals per capita	← log CPI Ratio	0.0020	0.0767 (.)
log CPI Ratio	← Arrivals per capita	0.0028	0.0267 (*)
Arrivals per capita	← log Freedom Index	0.0000	0.3400
log Freedom Index	← Arrivals per capita	0.0000	0.5633
Before COVID:			
Arrivals per capita	← Lagged GDP per capita	0.0078	< 0.0001 (***)
Lagged GDP per capita	← Arrivals per capita	0.0000	0.8233
Arrivals per capita	← log CPI Ratio	0.0010	0.1500
log CPI Ratio	← Arrivals per capita	0.0036	0.0167 (*)
Arrivals per capita	← log Freedom Index	0.0006	0.2333
log Freedom Index	← Arrivals per capita	0.0000	0.9000
After COVID:			
Arrivals per capita	← Lagged GDP per capita	0.0263	< 0.0001 (***)
Lagged GDP per capita	← Arrivals per capita	0.0000	0.6300
Arrivals per capita	← log CPI Ratio	0.0017	0.2967
log CPI Ratio	← Arrivals per capita	0.0022	0.1933
Arrivals per capita	← log Freedom Index	0.0000	0.7067
log Freedom Index	← Arrivals per capita	0.0000	0.9533

Notes: Significance is based on p-values: <0.001 (***), <0.01 (**), <0.05 (*), and <0.1 (.).

Arrows indicate the direction of information flow.

Appendix: Regression Output

Table 3: Translog regression of inbound tourist arrivals on economic and freedom indicators (with lagged Freedom Indicator)

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	12.7195	7.3393	1.73	0.0832	.
Time	-0.0234	0.0599	-0.39	0.6960	
Time ² /2	0.0044	0.0011	3.98	0.0001	***
GDP	1.2376	0.1922	6.44	< 0.001	***
FI	-8.2390	3.4136	-2.41	0.0159	*
CPIRatio	-5.2925	3.7098	-1.43	0.1538	
FI x Time	-0.0051	0.0123	-0.41	0.6799	
GDP x Time	-0.0007	0.0014	-0.51	0.6125	
CPIRatio x Time	0.0374	0.0389	0.96	0.3364	
FI ² /2	2.5475	0.8363	3.05	0.0023	**
GDP ² /2	0.0338	0.0040	8.55	< 0.001	***
CPIRatio ² /2	0.2005	0.3880	0.52	0.6054	
(FI x CPIRatio)/2	2.4704	1.7903	1.38	0.1677	
(GDP x CPIRatio)/2	-0.1207	0.1003	-1.20	0.2289	
(GDP x FI)/2	-0.6462	0.0979	-6.60	< 0.001	***
Tourism Interest	0.0548	0.0016	35.01	< 0.001	***
(Continent FE)Asia	-0.0120	0.0291	-0.41	0.6791	
(Continent FE)Europe	0.2080	0.0302	6.88	< 0.001	***
(Continent FE)North America	0.3687	0.0318	11.60	< 0.001	***
(Continent FE)Oceania	0.0009	0.0392	0.02	0.9814	
(Continent FE)South America	-0.1801	0.0385	-4.68	< 0.001	***
COVID FE	-0.4348	0.0429	-10.14	< 0.001	***

Notes:

GDP: Logarithm of GDP per capita of the country in the previous year;

CPIRatio: Logarithm of the ratio of consumer price index in a year to its value in the previous year;

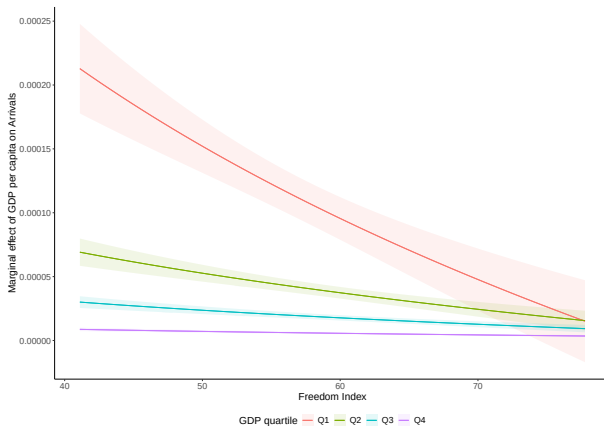
FI: Freedom index, as defined in (A.1);

Tourism Interest: Inbound tourist arrivals in 2005, proxying for inherent tourism potential;

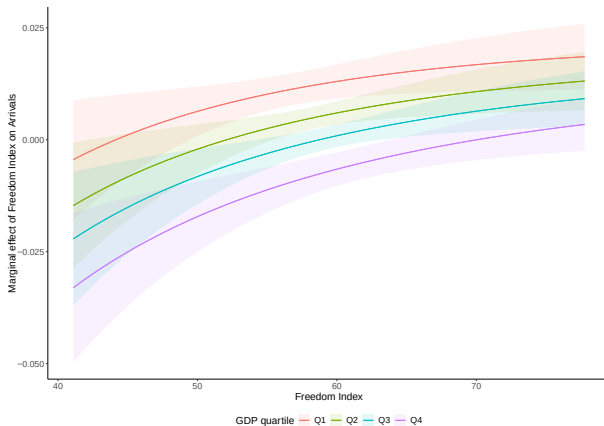
Continent FE: Continent-wise fixed effects;

COVID FE: Fixed effects for the COVID-19 pandemic (2020–2021).

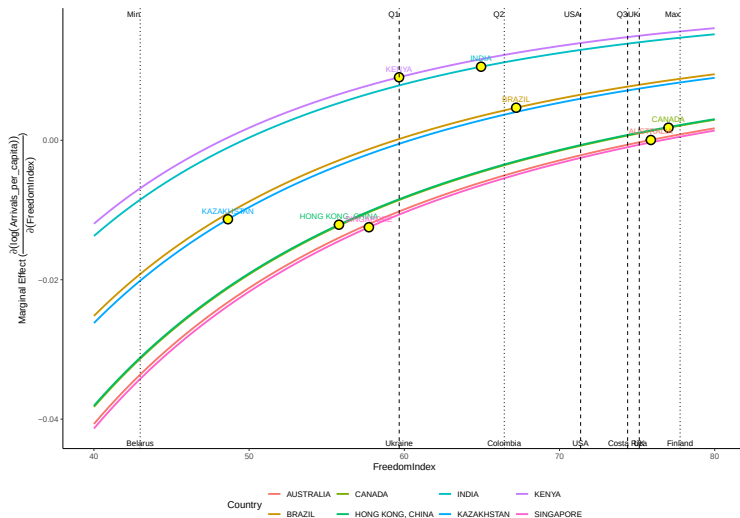
Appendix: Marginal Effect of ECO by CIVIL



Appendix: Marginal Effect of CIVIL



Appendix: Country-Level CIVIL Effects



Appendix: Substitutability Differs Across Countries

Table 5: Elasticity of Substitution (EoS) and Marginal Effect of Freedom Index (ME(F)) by Country and Year

	Finland		Sweden		Iceland		Croatia		Bangladesh		India	
Year	EoS	ME(F)	EoS	ME(F)	EoS	ME(F)	EoS	ME(F)	EoS	ME(F)	EoS	ME(F)
2007	3.738	0.003	2.671	0.003	3.284	0.003	-0.449	0.007	-0.087	0.016	-0.474	0.017
2008	3.611	0.003	2.663	0.003	-31.422	0.003	-0.307	0.007	-0.064	0.015	-0.471	0.017
2009	2.051	0.003	1.853	0.003	14.703	0.003	-0.666	0.006	-0.006	0.012	-0.514	0.017
2010	2.600	0.003	2.162	0.003	2.821	0.003	-0.811	0.006	-0.019	0.014	-0.484	0.017
2011	2.953	0.003	2.170	0.003	2.233	0.003	-0.494	0.007	-0.129	0.015	-0.354	0.016
2012	2.399	0.003	1.748	0.002	2.265	0.003	-0.453	0.007	-0.065	0.015	-0.334	0.016
2013	2.085	0.003	1.639	0.002	1.932	0.002	-0.562	0.007	-0.036	0.014	-0.340	0.016
2014	1.965	0.003	1.578	0.002	1.680	0.002	-0.981	0.006	-0.004	0.014	-0.217	0.015
2015	1.761	0.003	1.484	0.002	1.591	0.002	-1.166	0.006	0.016	0.013	-0.174	0.015
2016	1.747	0.003	1.466	0.002	1.472	0.002	-1.919	0.006	0.002	0.012	-0.159	0.015
2017	1.675	0.003	1.501	0.002	1.431	0.002	-1.274	0.006	-0.021	0.011	-0.090	0.014
2018	1.600	0.002	1.461	0.002	1.343	0.001	-1.746	0.006	-0.082	0.010	-0.075	0.014
2019	1.538	0.002	1.415	0.002	1.278	0.001	-5.581	0.005	-0.197	0.009	-0.054	0.013
2020	1.448	0.002	1.334	0.002	1.244	0.000	10.359	0.005	-0.627	0.008	-0.047	0.013
2021	1.544	0.002	1.390	0.002	1.311	0.001	-3.846	0.005	-1.246	0.007	-0.010	0.013
2022	-	-	1.621	0.002	-	-	-1.351	0.005	-	-	0.013	0.012
2023	1.700	0.002	-	-	-	-	-	-	-	-	0.011	0.011
2024	1.396	0.002	-	-	-	-	-	-	-	-	-0.004	0.011

Appendix: Elasticity of substitution vs features of countries in 2024

